

LLM Agents for Interlinked and Personalised Knowledge Bases



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This thesis evaluates whether LLM-generated structured notes and semantic links improve academic-paper retrieval and support a personalised Obsidian knowledge base.

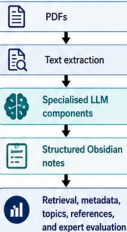
Motivation & Pipeline

Motivation

- Academic papers often remain isolated and difficult to rediscover.
- LLM components can generate structured notes, metadata, topic labels, and links.
- This thesis evaluates retrieval quality and extraction accuracy.

Pipeline

Modular multi-stage LLM pipeline



Research Questions

- RQ1: Retrieval performance
- RQ2: Metadata and topic extraction
- RQ3: Bibliography extraction
- RQ4: Expert evaluation of generated notes

RQ1 — Retrieval Performance

Flat Configuration



Raw PDF chunk retrieval

Structured Configuration



Generated-note retrieval

Link-Expanded Configuration



One-hop semantic-link score boost



Link-Expansion Achieved Highest Recall

One-hop semantic-link expansion produced the highest observed retrieval coverage, though overall performance gains were modest.

Configuration	Recall@5	Precision@5	MRR@5
Flat	0.6958	0.2350	0.7292
Structured	0.6958	0.2250	0.7438
Link-Expanded	0.7188	0.2300	0.7425



Best Recall@5:
Link-Expanded
(0.7188)



Best Precision@5:
Flat
(0.2350)



Best MRR@5:
Structured
(0.7438)



Structured notes did not consistently outperform raw PDF chunks. One-hop semantic-link expansion achieved the highest observed recall.

Extraction Accuracy

RQ2 — Metadata Extraction

Title exact match:	0.9750
Year exact match:	0.7750
Venue fuzzy match:	0.5000
Author F1-score:	0.8250

RQ2 — Topic Alignment

Strict lexical topic F1-score:	0.0957
Canonicalised topic F1-score:	0.3831

These topic evaluations use different procedures and should not be interpreted as a direct before-and-after improvement.

RQ3 — Bibliography Extraction

Precision:	Recall:	F1-score:
0.8433	0.5512	0.6667

Extracted references were usually valid, but many bibliography entries were omitted.

Key Conclusions

- Structured notes did not consistently outperform raw PDF chunks.
- Semantic-link expansion achieved the highest observed recall.
- Metadata extraction was stronger than topic-label alignment.
- Bibliography extraction showed high precision but moderate recall.

Limitations

- Small corpus and query set
- One personalised expert vocabulary
- No statistical uncertainty analysis

RQ4 — Expert Evaluation

Generated notes were assessed for:

- factual faithfulness
- contribution coverage
- readability
- utility for rediscovery